HEEJUN KIM

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EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) *September 2026 - present*
M.S. in Artificial Intelligence

Gwangju Institute of Science and Technology (GIST) *February 2020 - August 2026*
B.S. in Electrical Engineering and Computer Science *Military service leave (2 years)*
GPA: 4.2/4.5 (magna cum laude)

University of California, Berkeley (UC Berkeley) *June 2024 - August 2024*
Summer Session, GPA 4.0/4.0

RESEARCH EXPERIENCE

Scalable and Efficient AI Lab *September 2026 - present*
Korea Advanced Institute of Science and Technology, Advisor: Prof. Sehoon Kim *Seoul, South Korea*
· Improving reasoning capabilities and accelerating inference in recursive architectures

Learning and Adaptation Algorithm Lab *January 2026 - August 2026*
Seoul National University, Advisor: Prof. Taesup Kim *Seoul, South Korea*
· Conducting research on Agentic Memory Using LLM Uncertainty Estimation
· Studied internal behavioral mechanisms of LLMs underlying reasoning, hallucination, and memory recall

Data Science Lab *Feb 2025 - August 2026, January 2024 - June 2024*
Gwangju Institute of Science and Technology, Advisor: Prof. Sundong Kim *Gwangju, South Korea*
· Researched test-time adaptation with reinforcement learning for self-improving reasoning capabilities of large language models on the Abstraction and Reasoning Corpus (ARC)
· Investigated metacognition-driven strategies for helping large language models escape local minima in reasoning

Forschungszentrum Jülich *September 2024 - February 2025*
Korea-Germany Junior Research Fellowship Program 2024 *Jülich, Germany*
· Host Institute: INM-7 directed by *Prof. Simon B. Eickhoff* and contributed to Mathematical Neuroscience Group led by *Dr. Oleksandr Popovych*.
· Developed preprocessing pipeline for multi-modal MRI human connectome data using HPC and HyperThreading (Project Supervisor: *Dr. Kyesam Jung*)

PUBLICATIONS

*Equal contribution. †Corresponding author.

ResidualQuant: KV Cache Quantization for Looped Transformers with 2-Bit Residuals
Heejun Kim, Junyoung Lee, SangLyul Cho, Dongsu Han, Insu Han, Sehoon Kim
Under review. [Paper] [Project page]

From Noise to Diversity: Random Embedding Injection in LLM Reasoning
Heejun Kim*, Seungpil Lee*, Jewon Yeom, Jaewon Sok, Seonghyeon Park, Jeongjae Park, Taesup Kim†, Sundong Kim†
NeurIPS 2026, ICML 2026 Mechanistic Interpretability Workshop (Spotlight, 2.8% acceptance rate). [Paper] [Project page]

Hallucination as Commitment Failure: Larger LLMs Misfire Despite Knowing the Answer
Jewon Yeom, Jaewon Sok, Heejun Kim, Seonghyeon Park, Jeongjae Park, Taesup Kim
NeurIPS 2026, ICML 2026 Mechanistic Interpretability Workshop. [Paper]

On the Role of Proposal Support in Diffusion-Based Offline RL for Sequential Decision-Making
Yunho Kim, Jaehyun Park, Heejun Kim, Sejin Kim, Byung-Jun Lee, Sundong Kim
ICML 2026 DEMO Workshop. [Paper]

PROJECTS

Skill-based Reinforcement Learning with UCB Skill Selection *May 2025 - June 2025*

· Reproduced BeCL framework for unsupervised skill discovery in reinforcement learning environments

· Developed Multi-Armed Bandit approach with UCB algorithm to improve downstream task adaptation efficiency. [Poster]

Hyundai AI Challenge *September 2023 - October 2023*

· Processed large-scale port logistics datasets (cleaning, feature extraction, temporal aggregation).

· Built tree-based boosting models to predict ship waiting times.

· Ranked Top 25/330 teams among 1,300+ participants.

LEADERSHIP

Teaching Assistant *Fall 2025*

· Machine Learning & Deep Learning (AI5213/EC4213): led TA sessions, held office hours, managed assignments/exams, and organized Kaggle competition. [Homepage]

Korea Army Instructor *February 2021 - August 2022*

· Served as an Army instructor in the 50th Division in Daegu, South Korea

· Conducted basic training for approximately 2000 military personnel

EXTRA ACTIVITIES

Google Developer Student Club *September 2023 - February 2024*

· Participated in an AI study focused on machine learning and deep learning

· Studied models in ML/DL and analysis of top team models in Kaggle AI competitions

· Established study directions and organized study materials as a team leader

Financial Investment Conference *March 2023 - March 2024*

· Conducted a study on economics

· Analyzed and presented economic markets on a weekly basis

GIST Club Federation *February 2020 - December 2020*

· Managed clubs in Gwangju Institute of Science and Technology

SKILLS

Programming Languages	Python, C/C++
Python Packages & Framework	PyTorch, VeRL, Scikit-learn, Pandas, Numpy, Matplotlib, Ray
Operating System	Linux, Windows, MacOS
Others	Docker, Wandb, SLURM, HTCondor, SQLite

HONORS

Government Scholarship, Gwangju Institute of Science and Technology *2020 - August 2026*

Government STEM Scholarship *2020 - August 2026*

Dean's List, College of Information and Computing, GIST *Spring 2025, Fall 2025*

GIST International Internship Support Program *September 2024 - February 2025*

Korea-Germany Junior Research Fellowship Program 2024 *September 2024 - February 2025*

· Organized by the Max Planck POSTECH/KOREA Research (MPK)

· Fund: 12,000,000 KRW (8,197 EUR) by National Research Foundation of Korea (NRF), No. RS-2023-00254343.